A Project Report on

Live Emission Monitoring System for Petrol & Diesel Engines using IoT

In partial fulfilment of the requirements for the award of degree of

Bachelor of Technology in Mechanical Engineering

Submitted By

Sagar Bhagwat Patil **PRN:220101091013**

Vinay Mahendra Patil **PRN:230101092007**

Sandesh Dipak Bhamare **PRN:230101092005**

Under the Guidance of

Asst. Prof. Durgesh Borse

Department of Mechanical Engineering,

School of Engineering & Technology,

Sandip University, Nashik

(Academic Year- 2025-2026)

SANDIP UNIVERSITY, NASHIK

SCHOOL OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF MECHANICAL ENGINEERING

CERTIFICATE

This is to certify that the project work entitled **"Live Emission Monitoring System for Petrol & Diesel Engines using IoT"** submitted by

Sagar Bhagwat Patil	**PRN:220101091013**
Vinay Mahendra Patil	**PRN:230101092007**
Sandesh Dipak Bhamare	**PRN:230101092005**

and is submitted to Sandip University, Nashik in partial fulfilment of the requirements for the award of degree of Bachelor of Technology in Mechanical Engineering during the academic year 2025-2026.

Asst. Prof. Durgesh Borse
Project Guide

Mr. Pratap Garudkar,
Asst. Professor
External Examiner

Prof. Dr. Vishal N. Sulakhe
Head, Department of Mechanical Engineering

Prof. Dr. P. G. Burade
Associate Dean, SOET,
Sandip University, Nashik

ACKNOWLEDGEMENT

We take this opportunity with a great pleasure to express our sincere regards and deep sense of gratitude to our guide **Mr. Durgesh Borse**, Assistant Professor in Department Mechanical Engineering, SOET, Sandip University, Nashik for his valuable guidance, practical suggestions and encouragement to bring about the completion of project,

We are also thankful to all faculty of Mechanical Engineering Department of SOET, Sandip University, Nashik who have always co-operated while carrying out the project work.

We also express gratitude towards **Prof. Dr. Vishal N. Sulakhe,** Head of Mechanical Engineering Department, and **Prof. Dr. P. G. Burade**, Associate Dean, SOET, Sandip University, Nashik for their encouragement and timely suggestions.

Finally, we would like to thank our well-wishers, critics who helped directly or indirectly in the completion of this work.

Student Name	**PRN No.**	**Signature**
Sagar Bhagwat Patil	220101091013	______________
Vinay Mahendra Patil	230101092007	______________
Sandesh Dipak Bhamare	230101092005	______________

ABSTRACT

The rapid increase in vehicular population has significantly contributed to rising air pollution levels, necessitating the development of efficient and continuous emission monitoring systems. Conventional emission testing methods, such as periodic inspection and maintenance programs, fail to provide real-time insights into the actual emission behavior of vehicles during regular operation. This project presents the design and development of an IoT-based real-time vehicle emission monitoring system using multi-sensor integration to address these limitations.

The proposed system utilizes an ESP32 microcontroller as the central processing unit, integrated with MQ-7 and MQ-135 gas sensors for detecting carbon monoxide (CO) and nitrogen oxides (NOx), respectively, along with a particulate matter sensor for measuring PM2.5 and PM10 levels. To ensure sensor reliability and longevity, an exhaust gas sampling mechanism is employed, consisting of a stainless-steel probe, heat-resistant tubing, and a moisture trap to condition the exhaust gases before reaching the sensors.

The system continuously acquires sensor data, processes it in real time, and compares it against predefined emission thresholds. When pollutant levels exceed permissible limits, immediate alerts are generated through visual and audio indicators, while the data can also be transmitted to an IoT platform for remote monitoring and analysis. The proposed system emphasizes low-cost implementation, modular design, and energy-efficient operation, making it suitable for both research and practical deployment.

Experimental validation demonstrates stable sensor performance and reliable detection of emission variations under controlled conditions. The developed framework provides a scalable solution for continuous emission monitoring, contributing to improved environmental awareness, regulatory compliance, and sustainable transportation practices.

Keywords: IoT, Vehicle Emission Monitoring, ESP32, Gas Sensors, Real-Time Monitoring, Air Pollution Control.

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List of Abbreviations

ABBREVIATION	FULL FORM
ESP32	Embedded System Processing Unit
CO	Carbon Monoxide
Nox	Nitrogen Oxides
PM	Particulate Matter
OLED	Organic Light Emitting Diode
IoT	Internet of Things
Wi-Fi	Wireless Fidelity
LEMS	Live Emission Monitoring System
ADC	Analog to Digital Converter
PUC	Pollution Under Control

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

The increasing density of road transportation has intensified concerns regarding vehicular emissions, highlighting a critical gap in existing environmental monitoring infrastructure: the absence of continuous, real-time emission assessment during daily vehicle operation. Conventional emission testing methods rely on periodic inspections, offering only passive evaluation and failing to identify vehicles that exceed permissible pollution limits between testing intervals. The Live Emission Monitoring System (LEMS) addresses this limitation by integrating embedded sensing and IoT-based data acquisition technologies into a compact monitoring platform designed for continuous exhaust analysis. The system is built around the ESP32 microcontroller, utilizing MQ-series gas sensors and particulate matter sensing modules for accurate detection of harmful pollutants such as carbon monoxide (CO), nitrogen oxides (NOx), and airborne particulate matter. By employing a remote exhaust sampling probe and conditioned gas pathway, the design ensures sensor protection while enabling reliable real-time measurements under practical operating conditions. The system additionally focuses on energy efficiency and portability through optimized power regulation, while local visualization is provided via a low-power OLED interface. Through intelligent monitoring, alert mechanisms, and future cloud connectivity, the project delivers a comprehensive real-time emission analysis solution aimed at supporting environmental regulation and reducing pollution-related risks in modern transportation.

The transition from periodic emission testing toward continuous environmental monitoring is increasingly recognized as essential in modern transportation research, driven by growing urban air quality concerns and stricter regulatory frameworks. While existing pollution control systems provide baseline certification, the core challenge lies in leveraging Internet of Things (IoT) technologies to enable reliable, autonomous, and scalable emission surveillance. Early low-cost monitoring systems often suffered from inconsistent sensing accuracy, thermal instability, and insufficient protection against harsh exhaust environments. To overcome these limitations, robust emission monitoring requires multi-sensor fusion capable of correlating gas concentration data with particulate density measurements, thereby improving reliability and reducing false interpretations caused by transient engine conditions. The integration of MQ-7 and MQ-135 sensors enables detection of key gaseous pollutants, while a dedicated particulate sensor provides additional insight into diesel-related emissions, creating a comprehensive sensing architecture. Furthermore, operational viability depends heavily on efficient power management and stable voltage regulation, especially when deployed in mobile vehicular scenarios where fluctuations can affect sensor calibration and microcontroller performance. The system therefore incorporates regulated power conversion and modular hardware design to ensure consistent operation during prolonged monitoring sessions. Effective monitoring also requires secure data visualization and scalable communication pathways; local OLED-based display ensures immediate readability, while IoT-enabled transmission allows remote data logging and analysis regardless of location.

1.2 Scope of Project

The Scope of the proposed Live Emission Monitoring System (LEMS) arises from its ability to address several critical limitations present in conventional vehicular pollution monitoring practices, positioning it as an essential development in modern environmental safety technology. Traditional emission control methods rely heavily on periodic inspection frameworks such as Pollution Under Control (PUC) testing, which fail to provide continuous monitoring and therefore cannot identify vehicles exceeding emission thresholds during regular operation. Furthermore, existing low-cost monitoring attempts often suffer from inaccurate sensing, inadequate environmental protection for sensors, and poor integration of multiple pollutant measurements, leading to unreliable results and limited practical usability. The proposed LEMS directly addresses these gaps by combining embedded sensing technology, real-time data processing, and modular system design into a unified platform.

The system employs the ESP32 microcontroller as a computational core, enabling efficient realtime processing of sensor outputs while maintaining low power consumption. Data collected from MQ-7 and MQ-135 gas sensors, along with a particulate matter sensing module, are fused to provide a more comprehensive understanding of exhaust emissions compared to single-sensor approaches. This integrated sensing approach is important because previous systems frequently relied on isolated measurements that could not effectively represent real-world emission behavior. The inclusion of a controlled exhaust gas sampling probe ensures that sensors are protected from extreme temperatures and contamination while maintaining stable and repeatable measurements. Additionally, the system introduces a practical power architecture and modular design that supports long-term monitoring and future IoT integration. Real-time visualization through an OLED interface enhances usability by providing immediate feedback to users, while threshold-based alert functionality encourages proactive maintenance and pollution awareness.

The relevance of this project extends across multiple domains:

- **Environmental Protection and Public Health:** Enables continuous tracking of harmful exhaust emissions, supporting efforts to reduce air pollution and improve urban air quality.
- **Technological Advancement in IoT Monitoring:** Demonstrates efficient integration of gas sensing, particulate analysis, and embedded processing within a low-cost IoT framework suitable for educational and practical deployment.
- **Practical Vehicle Monitoring:** Provides real-time emission insights that support timely maintenance decisions and improve vehicle operating efficiency.
- **Foundation for Smart Environmental Systems:** Establishes a scalable platform that can later integrate cloud analytics, predictive models, or AI-driven emission pattern analysis.

The system therefore represents a significant step toward intelligent, data-driven emission monitoring and sustainable transportation practices.

1.3 PROBLEM STATEMENT

Conventional vehicle emission testing methods such as periodic Pollution Under Control (PUC) checks do not provide continuous real-time monitoring of vehicle exhaust emissions. Vehicles may emit excessive pollutants between testing intervals without detection, contributing significantly to environmental pollution and public health risks. Existing low-cost systems often lack integrated monitoring, alert mechanisms, and cloud-based data logging. Therefore, a real-time, portable, and low-cost emission monitoring system is required.

1.4 OBJECTIVES

The project objective focuses on the complete design and development of an integrated, realtime emission monitoring platform centered around the ESP32 microcontroller. Acting as the brain of the system, the ESP32 interfaces with multiple sensing modules and performs continuous processing of environmental data obtained from vehicle exhaust emissions. The primary sensing mechanism includes MQ-series gas sensors for carbon monoxide and nitrogen oxide detection, along with a particulate matter sensor capable of measuring PM2.5 and PM10 concentrations commonly associated with combustion engines.

To ensure accurate and safe operation, an exhaust gas sampling probe is employed to collect emission samples from the vehicle tailpipe and transport them through a conditioned pathway consisting of heat-resistant tubing and moisture filtration. This approach is critical because direct placement of sensors near the silencer can lead to thermal damage, contamination, and unstable readings. The sensor outputs are continuously acquired by the ESP32 through analog and serial interfaces, where custom-developed logic processes the incoming values and compares them against predefined emission thresholds. This processing method allows the system to differentiate normal operating variations from potentially harmful emission conditions, thereby improving reliability and reducing false alerts.

The primary functionality of the implemented system involves real-time monitoring and immediate user feedback. When pollutant concentrations exceed acceptable limits, the system activates visual and audible alerts through an LED and buzzer, enabling the operator to recognize excessive emissions instantly. Simultaneously, current values are displayed on a compact OLED screen, ensuring convenient local monitoring during operation. Future extensions include IoTbased cloud communication to support remote data logging and analysis.

In addition to sensing and monitoring functions, the project emphasizes operational stability and power reliability. The system utilizes regulated power conversion to maintain stable voltage levels for sensors and the microcontroller, preventing fluctuations that could impact measurement accuracy. The modular architecture also ensures ease of maintenance and scalability, allowing additional sensors or communication modules to be integrated without redesigning the core system.

The final platform demonstrates the feasibility of creating a low-cost, portable, and intelligent emission monitoring device capable of continuous operation under practical vehicular conditions. The high level of hardware and software integration confirms the potential of embedded IoT technologies in addressing modern environmental challenges.

CHAPTER 2

LITERATURE SURVEY

2.1 INTRODUCTION

The increasing concern regarding vehicular air pollution has created a strong demand for advanced emission monitoring systems capable of providing continuous and real-time environmental assessment. Conventional vehicle emission monitoring methods mainly rely on periodic inspection systems such as Pollution Under Control (PUC) testing, which provide only temporary evaluation of vehicle exhaust conditions. These approaches fail to detect excessive emissions occurring between scheduled inspections, allowing highly polluting vehicles to continue operating without immediate corrective action. As urban transportation density increases, the limitations of traditional monitoring approaches become more significant, creating the need for intelligent and automated monitoring solutions.

Recent advancements in embedded systems, gas sensing technology, and Internet of Things (IoT) platforms have enabled the development of compact and low-cost real-time environmental monitoring systems. Gas sensors such as MQ-series sensors are widely used for monitoring harmful pollutants including Carbon Monoxide (CO), nitrogen oxide related emissions, and volatile gases, while particulate matter sensors improve detection of smoke and dust emissions commonly associated with combustion engines. The integration of ESP32-based processing platforms, local display interfaces, alert mechanisms, and cloud communication creates opportunities for continuous emission monitoring, immediate user awareness, and future remote environmental analytics. The following literature review examines existing technologies, methodologies, and research contributions relevant to the development of the proposed Live Emission Monitoring System.

2.2 LITERATURE REVIEW

[1] IoT-based Air Quality Monitoring

- Method: This study demonstrated continuous environmental monitoring by integrating gas sensors with cloud-based dashboards, enabling remote visualization of air quality data in real time.
- Technology: Gas sensors, IoT cloud platform, wireless communication.

- Limitations: The system focused only on ambient environmental conditions and was not designed to handle high-temperature or high-concentration exhaust gases from vehicles. • Significance to Project: Establishes the usefulness of cloud-based monitoring but highlights the necessity for a specialized vehicle exhaust monitoring architecture.

[2] Vehicle Pollution Monitoring System

- Method: Implemented an emission sensing prototype using MQ-series gas sensors connected to a microcontroller for basic pollution measurement.
- Technology: MQ sensors, microcontroller-based data acquisition.
- Limitations: The system lacked particulate matter sensing, resulting in incomplete analysis of diesel-related emissions.
- Significance to Project: Validates the use of low-cost gas sensors but emphasizes the importance of adding PM sensing for comprehensive emission monitoring.

[3] Embedded Gas Detection System

- Method: Presented a low-cost gas detection system using MQ-series sensors interfaced with an embedded controller for monitoring hazardous gases.
- Technology: MQ-series sensors, Arduino platform.
- Limitations: Sensor readings became unstable in high-temperature environments, reducing reliability for exhaust gas applications.
- Significance to Project: Demonstrates affordability but highlights the need for sampling protection mechanisms such as a remote probe setup.

[4] IoT Environmental Monitoring

- Method: Used Wi-Fi and cloud storage to transmit environmental data from distributed sensors for remote monitoring.
- Technology: IoT platform, wireless communication modules.
- Limitations: The solution remained general-purpose and lacked focus on mobile vehicle-based monitoring requirements.
- Significance to Project: Provides background on remote monitoring but reinforces the need for a system specifically designed for vehicular emissions.

[5] Portable Air Pollution Detector

• Method: Developed a portable pollution detector that measured air quality and displayed values using onboard display modules.

• Technology: Gas sensors, local display interface.

• Limitations: The system lacked threshold-based alert mechanisms, limiting user awareness during critical pollution conditions.

• Significance to Project: Supports portability concepts while emphasizing the importance of real-time alert features in the proposed system.

[6] PM Monitoring System

• Method: Focused on particulate matter monitoring using dedicated dust sensors and embedded controllers.

• Technology: PM sensors, microcontroller platform.

• Limitations: Gas sensor integration was absent, resulting in incomplete analysis of overall vehicular emissions.

• Significance to Project: Highlights the necessity of combining gas and particulate sensing within a unified architecture.

[7] Vehicle Emission Tracking System

• Method: Implemented GPS and IoT technologies to track emission-related data at the vehicle level and transmit it remotely.

• Technology: GPS modules, IoT connectivity.

• Limitations: High system complexity and increased cost reduced suitability for low-cost educational applications.

• Significance to Project: Validates remote monitoring capability while motivating a simplified, affordable system design.

[8] Smart Emission Monitoring Prototype

• Method: Utilized gas sensors with an OLED display for immediate visualization of emission data.

• Technology: Gas sensors, OLED interface, microcontroller.

- Limitations: Sensors were directly exposed to harsh exhaust conditions without a protective sampling mechanism.
- Significance to Project: Demonstrates real-time display feasibility but emphasizes the need for sensor protection using sampling probes.

[9] IoT Pollution Alert System

- Method: Developed an alert system capable of sending pollution notifications through cloud infrastructure.
- Technology: Sensor network, IoT cloud alerts.
- Limitations: Lack of local display reduced usability when internet connectivity was unavailable.
- Significance to Project: Justifies integrating local OLED visualization alongside IoT features.

[10] Embedded Emission Analysis

- Method: Introduced sensor fusion approaches to improve interpretation of emissionrelated data collected from multiple sources.
- Technology: Multi-sensor integration, embedded processing.
- Limitations: No consideration was given to power optimization strategies for continuous monitoring systems.
- Significance to Project: Supports data fusion concepts while highlighting the necessity for efficient power design.

[11] Mobile Air Quality Node

- Method: Proposed a battery-powered IoT node for portable environmental monitoring applications.
- Technology: Battery-powered sensors, wireless IoT transmission.
- Limitations: Sensor drift and instability affected long-term reliability during extended operation.
- Significance to Project: Demonstrates portability but emphasizes calibration and stability improvements required in vehicular monitoring.

Table 2.1.1 Comparative Analysis of Existing Work

Paper No.	Reference / Focus	Key Facilities	Scope	Key Limitation
[1]	IoT-based Air Quality Monitoring	Gas sensors, cloud dashboard	Real-time ambient air monitoring	Not designed for vehicle exhaust monitoring
[2]	Vehicle Pollution Monitoring System	MQ sensors, microcontroller	Emission sensing prototype	No particulate matter measurement
[3]	Embedded Gas Detection System	MQ-series sensors, Arduino	Low-cost gas detection	Limited accuracy in high temperature environments
[4]	IoT Environmental Monitoring	Wi-Fi, cloud storage	Remote environmental data logging	Not specific to vehicles
[5]	Portable Air Pollution Detector	Gas sensors, display module	Portable pollution awareness	No real-time threshold alert system
[6]	PM Monitoring System	Dust sensor, microcontroller	PM2.5/PM10 measurement	No gas sensor integration
[7]	Vehicle Emission Tracking System	GPS + IoT	Vehicle-level tracking	High cost and complex architecture
[8]	Smart Emission Monitoring Prototype	Gas sensors, OLED display	Real-time display of emissions	No sampling probe for sensor safety
[9]	IoT Pollution Alert System	Cloud alerts, sensor network	Remote pollution alerts	Lacked local display interface
[10]	Embedded Emission Analysis	Sensor fusion approach	Improved data interpretation	No power optimization discussed
[11]	Mobile Air Quality Node	Battery-powered IoT	Portable monitoring	Sensor drift and instability
[12]	Proposed Project (2025)	ESP32, MQ-7, MQ-135, PM Sensor, OLED, Sampling Probe, IoT	Unified real-time emission monitoring for petrol & diesel vehicles	Stage–1 prototype; field calibration pending

2.3 EXISTING TECHNOLOGIES USED IN VEHICLE EMISSION MONITORING

- **MQ-Series Sensor Monitoring:**

MQ-series gas sensors such as MQ-7 and MQ-135 are widely used for low-cost environmental and emission analysis due to their sensitivity to harmful gases including carbon monoxide (CO), nitrogen oxides (NOx), and volatile compounds. Research indicates that combining multiple gas sensors improves detection reliability by enabling comparative analysis rather than relying on a single measurement source. Such sensor fusion enhances the ability to differentiate genuine emission spikes from temporary fluctuations caused by engine load changes or surrounding environmental conditions. This multi-sensor approach forms the basis for reliable real-time exhaust monitoring systems.

- **Particulate Matter (PM) Detection:**

Particulate sensors capable of measuring PM2.5 and PM10 concentrations are increasingly adopted in pollution monitoring studies, particularly for diesel emission analysis. Optical particle counting techniques allow real-time estimation of particulate concentration by detecting light scattering caused by airborne particles. Integrating PM sensing alongside gas measurement provides a comprehensive emission profile and improves overall monitoring accuracy.

- Real-Time Monitoring and Communication:

- **IoT Connectivity and Data Transmission:**

Research has established the importance of wireless communication frameworks in environmental monitoring systems. Wi-Fi-enabled microcontrollers such as ESP32 allow realtime data transmission to cloud dashboards for remote monitoring and analysis. While cloud platforms support data logging and long-term trend analysis, local processing remains essential for immediate response and threshold-based alerts. This dual approach ensures system usability even when network connectivity is limited or unavailable.

- Sampling and Sensor Protection Techniques:

➢ **Exhaust Gas Sampling Probe:**

Studies focusing on emission analysis emphasize the necessity of controlled gas sampling to protect sensitive sensors from extreme exhaust temperatures and contamination. The use of stainless-steel sampling probes combined with heat-resistant tubing enables safe transport of exhaust gases to a conditioned sensing chamber. Additional elements such as moisture filters and sampling pumps stabilize gas flow and improve measurement consistency, significantly increasing sensor lifespan and reducing measurement errors.

- Energy Management and Stability:

➢ **Power Regulation and Efficiency:**

A major area of research in portable monitoring systems focuses on maintaining stable operation under varying power conditions. Voltage regulation circuits and efficient power distribution are essential to prevent sensor drift and microcontroller instability. Proper power management ensures continuous monitoring capability and supports future expansion with IoT communication and additional sensing modules.

- Central Processing and Control:

➢ **ESP32 Embedded Platform:**

The ESP32 microcontroller is widely adopted for IoT-based embedded systems due to its balance between processing capability, low power consumption, and integrated wireless communication. The platform enables simultaneous sensor acquisition, data processing, display management, and communication handling. Its multi-interface support (ADC, UART, I^2C) allows seamless integration of gas sensors, particulate sensors, and display modules within a single architecture.

- User Interface and Feedback:

➢ **OLED Display and Alert Mechanisms:**

Research in embedded monitoring emphasizes the importance of instant user feedback to support real-time decision-making. OLED displays provide compact, low-power visualization of sensor values, while alert systems using LEDs and buzzers notify users when pollutant levels exceed predefined thresholds. This combination improves usability and ensures immediate awareness of harmful emission conditions.

2.4 RESEARCH GAP:

Although significant research has been carried out in environmental monitoring, several important challenges remain unresolved in the context of a unified vehicular emission monitoring platform:

- **Integration Challenge:**

Most existing systems focus either on gas monitoring or particulate analysis individually, lacking a unified architecture that combines multi-sensor detection, secure sampling, and real-time visualization into a single portable system suitable for vehicle exhaust environments.

- **Accuracy and Environmental Protection Trade-Off:**

Many low-cost sensor systems struggle with measurement reliability due to direct exposure to high temperatures and contaminants from exhaust gases. Protective sampling mechanisms are often absent, leading to reduced sensor lifespan and unstable readings.

- **Limited Real-World Validation:**

While numerous studies demonstrate successful laboratory testing, few provide practical validation under real vehicular operating conditions where vibration, temperature variation, and fluctuating emissions significantly affect performance.

- **Power and Scalability Constraints:**

Some IoT-based solutions emphasize cloud functionality but neglect efficient local processing and power stability, reducing reliability during continuous operation or mobile deployment.

- **Lack of User-Centric Interface:**

Existing prototypes frequently omit local displays or immediate alert systems, limiting usability when internet connectivity is unavailable or delayed.

The proposed system addresses these gaps by developing a modular and energy-efficient Live Emission Monitoring System that integrates MQ-series gas sensors, particulate monitoring, protected exhaust sampling, and real-time processing through the ESP32 platform. This unified approach aims to provide accurate, continuous emission monitoring while ensuring sensor safety, scalability, and practical deployment capability.

CHAPTER 3

METHODOLOGY

3.1 PROBLEM STATEMENT

The challenge in contemporary vehicular pollution control lies in overcoming the limitations of periodic emission monitoring systems and the absence of continuous real-time assessment during daily vehicle operation. Despite regulatory frameworks such as periodic Pollution Under Control (PUC) testing, a significant gap remains between scheduled inspections and real-world emission behavior, allowing highly polluting vehicles to operate unnoticed for extended periods. Existing technological solutions exhibit several major shortcomings that hinder effective environmental monitoring:

1. **Inaccurate and Incomplete Detection:** Many currently available systems rely on single gas sensors or basic air quality detectors, resulting in unreliable measurements, poor differentiation between emission variations, and incomplete pollutant analysis due to lack of particulate monitoring.
2. **Sensor Vulnerability and Environmental Limitations:** Gas sensors exposed directly to vehicle exhaust are prone to thermal damage, contamination, and rapid degradation, leading to unstable readings and reduced long-term reliability.
3. **Fragmented System Architecture:** Existing solutions often focus separately on sensing, display, or IoT communication without integrating sampling, processing, and alert mechanisms into a unified, modular framework.
4. **Lack of Real-Time Monitoring Capability:** Periodic testing methodologies fail to provide continuous monitoring, preventing early detection of high-emission conditions and reducing the effectiveness of corrective maintenance.
5. **Limited User Interaction:** Many systems rely solely on cloud dashboards without local displays or immediate alert mechanisms, limiting practical usability when network connectivity is unavailable.

3.2 PROPOSED SOLUTION

This project proposes the design and implementation of a Live Emission Monitoring System (LEMS) that integrates embedded sensing, real-time processing, and IoT-ready architecture to enable continuous monitoring of vehicle exhaust emissions. The system employs a multi-sensor approach to measure gaseous pollutants and particulate matter while ensuring sensor protection through a dedicated exhaust sampling mechanism.

The architecture is centered around the ESP32 microcontroller, which continuously acquires data from MQ-7 and MQ-135 gas sensors along with a particulate matter sensor. Exhaust gases are collected safely using a stainless-steel sampling probe and transferred through a conditioned path to a protected sensor chamber, ensuring measurement stability and extending sensor lifespan. The ESP32 processes sensor data in real time, compares values against predefined thresholds, and provides immediate feedback through local visualization and alert systems. Future integration with IoT platforms enables remote monitoring and long-term data analysis.

Energy stability and operational reliability are achieved through regulated power management, ensuring consistent performance under mobile vehicular conditions.

Additionally, the inclusion of a compact OLED display allows users to observe live emission levels without relying solely on remote dashboards, improving real-time awareness and usability.

The specific goals of this project include:

- Developing a multi-sensor monitoring architecture capable of measuring CO, NOx, and particulate matter simultaneously.
- Implementing a protected exhaust sampling mechanism to improve sensor reliability and measurement consistency.
- Integrating real-time local visualization and alert features for immediate user awareness.
- Providing a scalable IoT-ready framework for future remote monitoring and analytics.
- Creating a compact, modular prototype suitable for installation on both petrol and diesel vehicles.

3.3 OBJECTIVES

The primary objective of this project is to design and develop an intelligent, low-cost, and reliable emission monitoring platform capable of continuously analyzing vehicle exhaust pollutants and providing real-time feedback to users. By enabling continuous monitoring rather than periodic testing, the system aims to support early detection of excessive emissions and contribute to improved environmental safety.

This system will:

- Perform real-time monitoring of exhaust gases and particulate emissions using integrated sensor fusion.
- Ensure stable and reliable operation through protected sampling and efficient power regulation.
- Provide immediate local feedback through OLED display and threshold-based alert mechanisms.
- Establish a scalable foundation for IoT-enabled remote monitoring and data logging.
- Serve as a modular prototype for future smart environmental monitoring applications in transportation systems.

3.4 METHODOLOGY

The methodology adopted for the proposed Live Emission Monitoring System is as follows:

• Exhaust gases emitted from the vehicle are collected through a sampling inlet using a copper tube arrangement, which safely directs the exhaust sample toward the sensing chamber while reducing direct thermal exposure to the sensors.

• The sensing chamber contains MQ-7 sensor for Carbon Monoxide (CO) detection, MQ-135 sensor for monitoring harmful gaseous pollutants, and particulate matter sensor for measuring dust and smoke concentration.

• These sensors continuously monitor pollutant concentration levels and generate corresponding electrical output signals based on variations in gas and particulate presence.

• The ESP32 microcontroller receives the sensor signals and performs real-time processing of the collected emission data according to predefined threshold values.

• The processed emission readings are displayed on the OLED display unit, allowing real-time local monitoring of CO, NOx, and particulate matter levels.

• LED indicators and buzzer are incorporated to provide warning alerts based on emission conditions. Moderate emission levels activate visual indication, while high emission levels trigger both visual and audible alerts.

• The ESP32 utilizes Wi-Fi connectivity to periodically upload emission data to the ThingSpeak cloud platform for remote monitoring and data logging.

• The cloud platform enables real-time access to emission trends, graphical analysis, and storage of historical monitoring data.

• The integrated system provides a compact, portable, low-cost, and efficient real-time vehicle emission monitoring solution suitable for academic demonstration and environmental monitoring applications.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 SYSTEM ARCHITECTURE

The overall architecture of the Live Emission Monitoring System (LEMS) is based on a tightly integrated, modular design focused on real-time sensing, continuous data processing, and intelligent environmental monitoring. The system utilizes the ESP32 microcontroller as the central processing unit, integrating multiple specialized modules for emission sensing, sampling control, display visualization, and power management. The architecture is designed to provide continuous monitoring of vehicle exhaust gases while ensuring sensor safety and operational reliability.

Block Diagram

The block diagram illustrates the operational flow of the system, describing how various components interact to achieve the dual objectives of accurate emission measurement and realtime monitoring under practical vehicular conditions. The integration of these modules ensures that the system remains in a constant monitoring state, continuously sampling exhaust gases and processing sensor data, while automatically triggering alerts whenever emission thresholds are exceeded, ensuring reliability and immediate feedback.

- **Exhaust Sampling Probe (Gas Input):** This module acts as the primary sensing interface with the vehicle exhaust. A stainless-steel probe collects emission samples from the tailpipe and transports them through a conditioned pathway. This approach protects sensitive sensors from high temperatures and contamination while ensuring stable sampling for reliable measurements.

- **Gas Sensors (MQ-7 and MQ-135):** These modules serve as the primary gas detection units. MQ-7 measures carbon monoxide levels while MQ-135 detects nitrogen oxides and general air quality indicators. Their analog output signals are continuously read by the ESP32 and analyzed to identify pollutant concentration changes.

- **Particulate Matter Sensor (PM2.5/PM10):** This sensor provides real-time measurement of particulate emissions commonly associated with diesel combustion. The integration of PM sensing with gas monitoring enables comprehensive emission analysis rather than isolated pollutant detection.

- **ESP32 Microcontroller (Central Controller/Processing Unit):** The processing core of the system. It executes sensor data acquisition, performs threshold comparison, manages communication between modules, and controls alert logic. The ESP32 also handles display updates and prepares data for optional IoT transmission.

- **OLED Display and Alert Module:** This block provides user interaction and immediate feedback. The OLED screen displays live emission values, while LED and buzzer alerts are activated when pollutant levels exceed predefined thresholds, enabling instant awareness without reliance on remote dashboards. The modular integration of sensing, processing, and user feedback components guarantees continuous emission monitoring while maintaining system stability, sensor safety, and scalability for future IoT-based expansion.

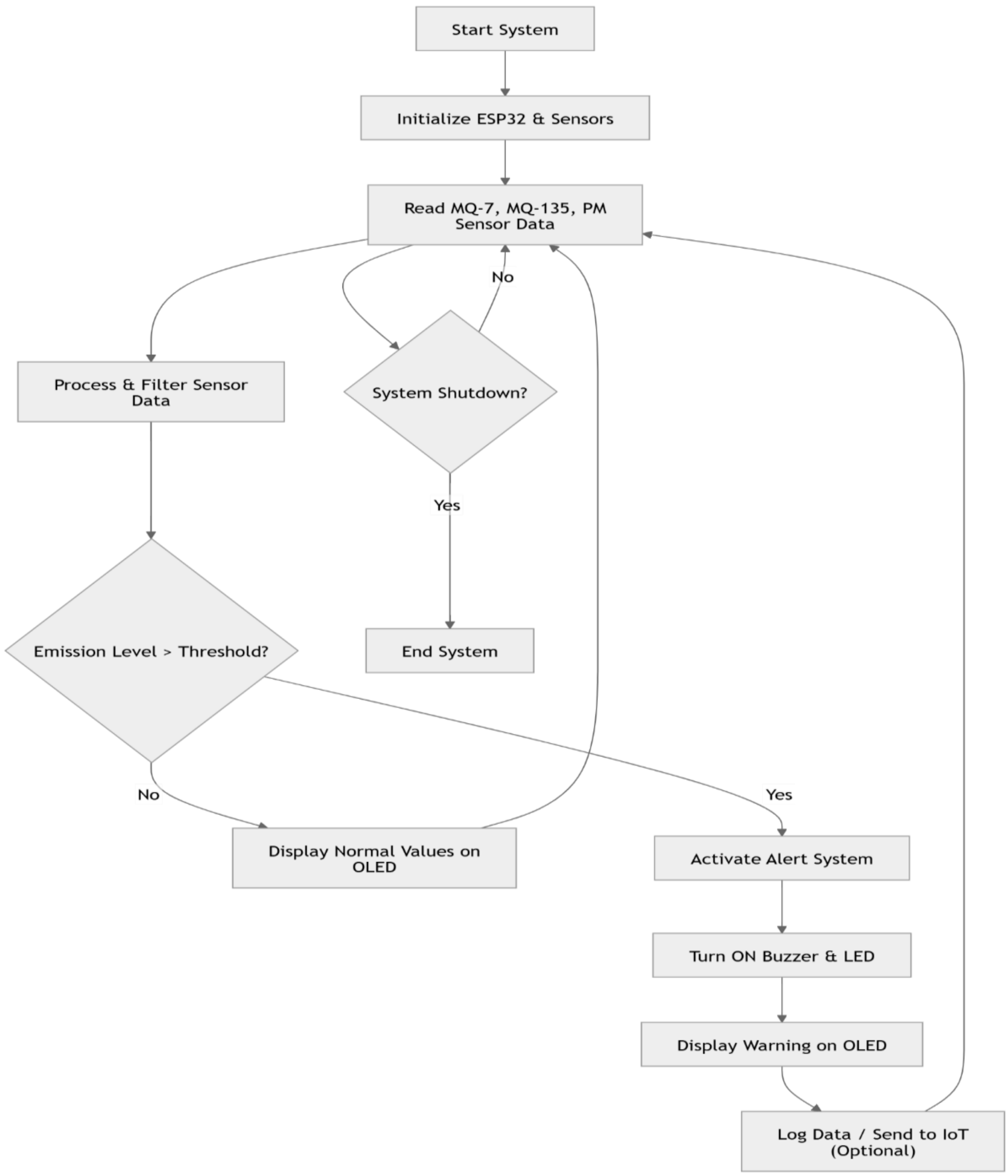

Fig.4.1.1 Block Diagram

This flowchart represents the working of the Live Emission Monitoring System for Petrol & Diesel Engines using IoT. The process starts with system initialization, where the ESP32 and sensors (MQ-7, MQ-135, and PM sensor) are activated. The system continuously reads emission data and processes it using filtering techniques. A decision is made by comparing emission levels with predefined threshold values. If the values are within limits, normal data is displayed on the OLED screen. If the threshold is exceeded, the alert system is activated, triggering a buzzer and LED, along with a warning display. The system can also log data or send it to an IoT platform for monitoring.

4.2 HARDWARE COMPONENTS

Process A: Live Emission Monitoring System Operational Sequence (System Workflow)

This flowchart illustrates the complete operational sequence and decision-making process of the Live Emission Monitoring System, focusing on real-time emission detection, processing, and alert generation using IoT-based architecture.

Start System & Initialization:

- Start: The system operation begins when power is supplied to the system or the vehicle ignition is turned ON.
- Initialize ESP32 & Sensors: The ESP32 microcontroller initializes all connected components, including MQ-7, MQ-135, and particulate matter sensors. Communication interfaces such as ADC and serial communication are configured. The OLED display is activated to show system status.

Core Monitoring State:

- Read Sensor Data (MQ-7, MQ-135, PM): The system continuously acquires real-time emission data from gas and particulate sensors. These sensors detect carbon monoxide, nitrogen oxides, and particulate matter levels from vehicle exhaust.

- Shutdown Command? The system checks whether a shutdown signal is received.

o If Yes: The system stops operation and transitions to the stop state. o

If No: The system continues monitoring and processing data.

- Filter and Process Data: Raw sensor readings are processed using filtering techniques to remove noise and improve accuracy. This ensures stable and reliable emission values.

Decision-Making Stage:

- Compare with Threshold: The processed sensor data is compared with predefined emission limits. This is the critical decision point of the system.
- If Below Limit:

The system continues normal monitoring. Real-time emission values are displayed on the OLED screen, indicating safe operating conditions. The system loops back to continuous sensing.

- If Above Limit:

The system detects excessive emissions and transitions to the alert stage.

Alert Execution:

- Trigger Alert: Once the emission exceeds the threshold, the system activates the alert mechanism.
- Activate Buzzer and LED: Audible and visual alerts are triggered to notify the user immediately.
- Display Warning Message: The OLED screen shows a warning message along with current emission values.
- Store or Send Data: The system logs the emission data or sends it to an IoT platform for remote monitoring and analysis.

End State:

- Stop System: If a shutdown command is received, the system safely terminates all operations.

Process B: Continuous Monitoring & User Feedback Flow

This process represents the real-time interaction between the system and the user. The system continuously monitors emission levels and provides instant feedback through the OLED display. Users are notified immediately when emission levels exceed safe limits through buzzer and LED alerts. The system ensures reliable performance by maintaining a continuous loop of sensing, processing, decision-making, and alerting, making it suitable for real-world vehicle emission monitoring applications.

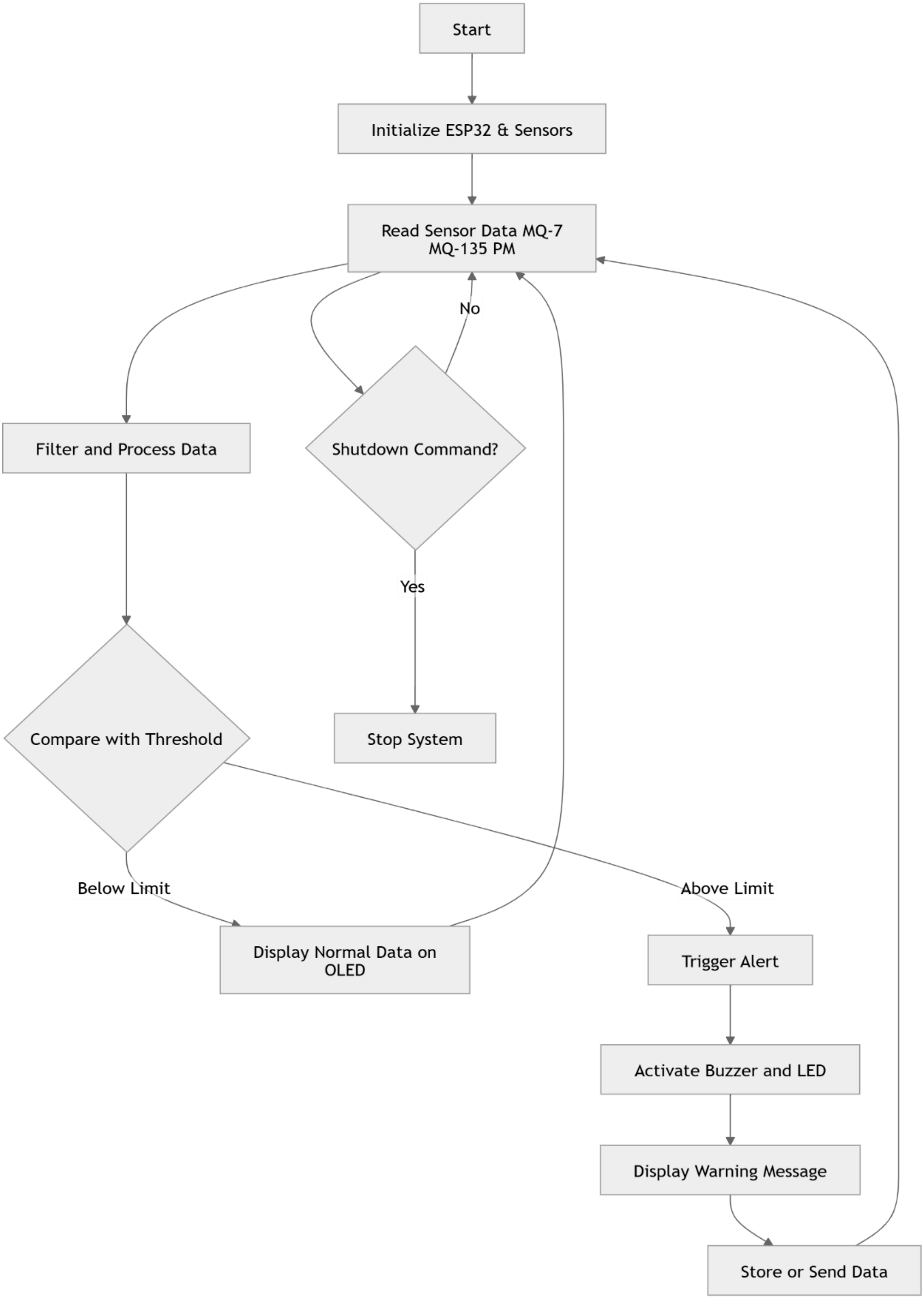

Figure 4.1.2 Emission Monitoring Alert Workflow

1. Start & Monitoring State:

• Start / System Initialization:

The workflow of the Live Emission Monitoring System begins when power is supplied to the system or the vehicle ignition is turned ON. At this stage, the ESP32 microcontroller initializes all connected hardware components, including the MQ-7 sensor for carbon monoxide detection, MQ-135 sensor for air quality and nitrogen oxide monitoring, and the particulate matter sensor for PM2.5 and PM10 measurement. Along with sensor initialization, the system configures communication interfaces such as analog-to-digital conversion channels and serial communication protocols. The OLED display is also activated to provide visual feedback to the user, indicating that the system is ready for operation. Proper initialization ensures that all components function synchronously and accurately throughout the monitoring process.

• Continuous Data Acquisition:

Once initialization is complete, the system transitions into the core monitoring state. In this phase, the ESP32 continuously collects real-time emission data from all connected sensors. This process occurs in a loop, ensuring uninterrupted monitoring of vehicle exhaust conditions. The system remains in this active state under normal operating conditions, where emission levels are within acceptable limits. The continuous data acquisition ensures that any sudden changes in emission levels are detected instantly, enabling timely response and system reliability.

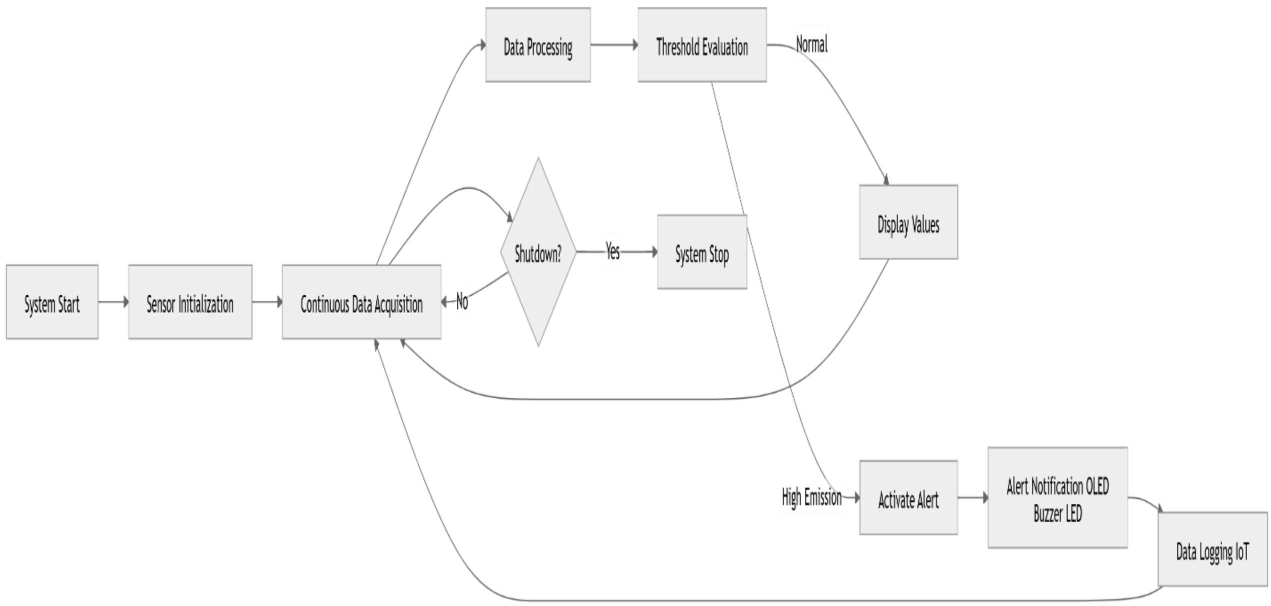

Figure 4.1.3 Interactive Communication Protocol Flow

2. Command Processing and Decision Stage:

• **Process Input (Sensor Data Evaluation):**

After acquiring raw data from sensors, the ESP32 processes these readings to improve accuracy and reliability. This includes applying filtering techniques to remove noise and fluctuations caused by environmental factors such as temperature, humidity, or engine vibration. The processed data represents the actual emission levels, which are then used for further analysis. This stage is crucial because accurate data processing directly impacts the effectiveness of emission monitoring and alert generation.

• **Shutdown Command Check:** Simultaneously, the system checks whether a shutdown command has been received from the user or an external control mechanism. o If Yes: The system immediately halts all operations and transitions to the stop state, ensuring safe shutdown of sensors and peripherals. o If No: The system continues its normal monitoring operation and proceeds to the next decision stage.

• **Threshold Comparison:**

The processed emission values are compared with predefined threshold limits set according to environmental standards. This comparison acts as the core decision-making point of the system. It determines whether the current emission levels are safe or if they exceed permissible limits, requiring immediate action.

3. Execution Paths:

Based on the outcome of the threshold comparison, the system follows one of the following execution paths:

A. Normal Monitoring Condition (Below Threshold):

• **Display Normal Data on OLED:**

If the emission levels are within safe limits, the system continues normal operation. The OLED display shows real-time values of pollutants such as CO, NOx, and particulate matter. This allows the user to continuously observe emission conditions and verify that the vehicle is operating within acceptable environmental limits.

• **Loop Back to Monitoring:**

After displaying the current data, the system loops back to the continuous data acquisition stage. This ensures that monitoring is ongoing and dynamic, allowing the system to respond instantly to any changes in emission levels.

B. Alert Condition (Above Threshold):

• Trigger Alert System:

If the emission levels exceed predefined thresholds, the system identifies a high emission condition and immediately initiates the alert mechanism.

• Activate Buzzer and LED:

The ESP32 activates both the buzzer and LED indicators to provide immediate audible and visual alerts to the user. This ensures that the user is quickly notified about unsafe emission levels, even without looking at the display.

Display Warning Message:

The OLED screen displays a warning message along with current emission values. This helps the user understand the severity of the situation and take necessary corrective actions, such as vehicle maintenance or reducing engine load.

• Store or Send Data:

Additionally, the system can store the emission data locally or transmit it to an IoT platform for remote monitoring and analysis. This feature enables long-term tracking of emission trends and supports data-driven decision-making.

• Return to Monitoring:

After the alert process is completed, the system automatically returns to the monitoring loop, ensuring continuous operation and further tracking of emission levels.

C. Shutdown Command:

• Execute System Shutdown:

When a shutdown command is detected, the ESP32 performs a controlled shutdown sequence. It stops data acquisition, deactivates sensors, and ensures that all ongoing processes are safely terminated.

• End / Power Save Mode:

Finally, the system enters a low-power or inactive state, conserving energy and ensuring system safety. It remains in this state until it is manually restarted or power is restored.

• SYSTEM PLATFORM

4.3 CIRCUIT CONNECTIONS

The successful development of the Live Emission Monitoring System (LEMS) depends on the integration of reliable embedded hardware and efficient software architecture optimized for realtime sensing, data processing, and environmental monitoring in vehicular conditions.

• **Hardware Requirements:**

1. ESP32 Microcontroller Module

• Function: Acts as the central processing unit of the system, responsible for acquiring sensor data, executing threshold analysis, controlling alert mechanisms, managing the OLED interface, and enabling optional IoT communication. Its integrated Wi-Fi capability supports future cloud connectivity while maintaining low power operation. • Key Libraries: Arduino IDE / ESP-IDF environment, WiFi library, Wire (I²C), and ADC reading functions for sensor interfacing.

Figure 4.3.1 ESP32 Development Module

2. MQ-7 Gas Sensor (Carbon Monoxide Detection)

• Function: Detects carbon monoxide (CO) present in vehicle exhaust gases. The sensor provides analog signals proportional to CO concentration, enabling real-time pollutant monitoring. • Key Libraries: Analog input processing through ESP32 ADC channels; optional filtering algorithms for signal stabilization.

Figure 4.3.2 MQ-7 Gas Sensor Module

3. MQ-135 Gas Sensor (NOx / Air Quality Detection)

• Function: Monitors nitrogen oxides and general air quality gases produced during combustion. This sensor complements MQ-7 readings for broader emission analysis.
• Key Libraries: ADC libraries and calibration functions for gas concentration approximation.

Figure 4.3.3 MQ-135 Gas Sensor Module

4. Particulate Matter Sensor (PM2.5 / PM10)

• Function: Measures particulate emissions commonly produced by diesel engines. It provides real-time particulate concentration data that enhances overall emission profiling.
• Key Libraries: Serial communication (UART) libraries for data acquisition.

Figure 4.3.4 PM Sensor Module

5. OLED Display (I²C Interface)

- Function: Provides real-time visualization of pollutant levels, system status, and alert notifications. Designed for low power consumption and clear readability.
- Key Libraries: Adafruit SSD1306 / U8g2 display libraries.

Software Requirements

The software architecture converts raw sensor data into meaningful emission insights through layered processing running on the ESP32 platform.

1. Development Environment: Arduino IDE or ESP-IDF is used for firmware development, offering stability and easy hardware integration.
2. Programming Language: Embedded C/C++ is used for real-time sensor acquisition, threshold analysis, and hardware control.
3. Sensor Interface Libraries: Analog input modules are used to read MQ sensor outputs, while UART libraries handle communication with the PM sensor.
4. Display Interface Libraries: OLED libraries are utilized to format and display live pollutant data and system status information.
5. Core Logic Script: A custom embedded program controls system states, continuously processes sensor values, performs threshold comparison, and manages alert sequences.
6. Optional IoT Communication: Wi-Fi libraries allow expansion toward cloud-based monitoring and remote data logging.
7. Hardware Control Libraries: GPIO functions manage buzzer and LED activation as part of the alert protocol.

4.4 SYSTEM SPECIFICATIONS

The system is defined by functional capabilities and measurable performance targets necessary for reliable real-time emission monitoring under practical vehicular conditions.

4.4.1 Functional Specifications

The Live Emission Monitoring System must perform five core functions autonomously

:

1. Real-Time Emission Monitoring: Continuous measurement of CO, NOx, and particulate matter emissions using integrated sensors.
2. Protected Sampling Mechanism: Safe exhaust gas collection through a probe and conditioning line to ensure sensor longevity and stable readings.
3. Immediate User Feedback: Real-time data visualization via OLED display and instant alert activation when thresholds are exceeded.
4. Modular Expansion Capability: Support for IoT communication and additional sensors without redesigning the core architecture.
5. Continuous Monitoring Operation: Ability to operate reliably during extended vehicle operation with stable power regulation.

4.4.2 Performance Specifications

The system is expected to meet measurable performance metrics to ensure dependable operation:

• Detection Reliability: Sensor readings should accurately reflect emission variations with stable operation after calibration.

- Response Time: Alert triggering should occur within seconds of threshold violation during monitoring.
- Sampling Accuracy: Exhaust sampling system should provide consistent gas flow while minimizing contamination effects.
- Operational Runtime: System should sustain continuous monitoring during typical driving durations using regulated power supply.
- Voltage Stability: Power circuitry must maintain stable voltage levels to prevent sensor drift or controller instability.

4.5 PROTOTYPE DEVELOPMENT

The hardware prototype of the proposed Live Emission Monitoring System was successfully developed using ESP32 microcontroller, MQ-7 sensor, MQ-135 sensor, DSM501B particulate sensor, OLED display, buzzer, vacuum pump, and exhaust gas sampling arrangement. A compact enclosure was used for safe integration of components and practical testing.

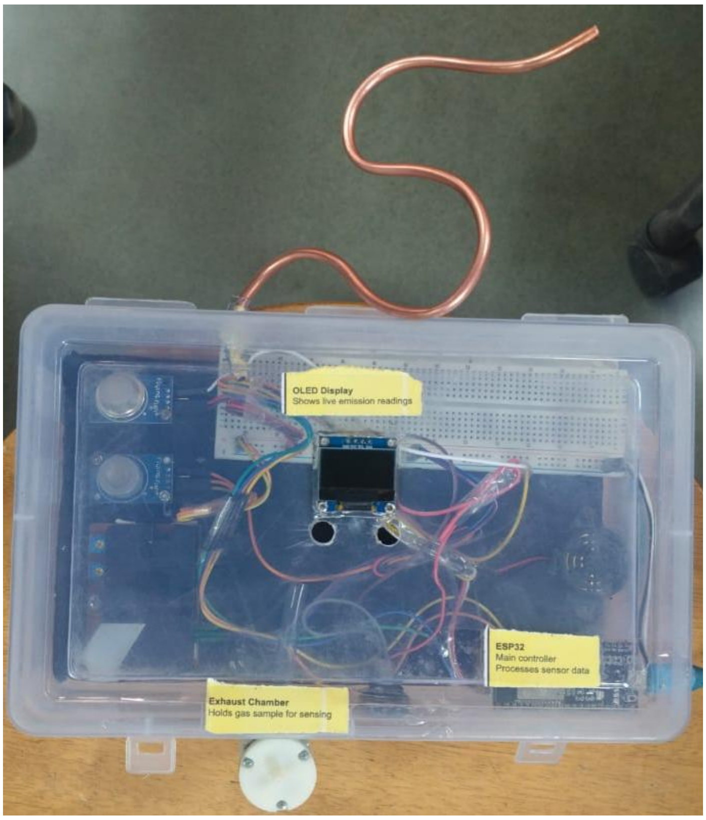

Figure 4.5.1 Developed Prototype

Figure 4.5.2 Sensor Integration

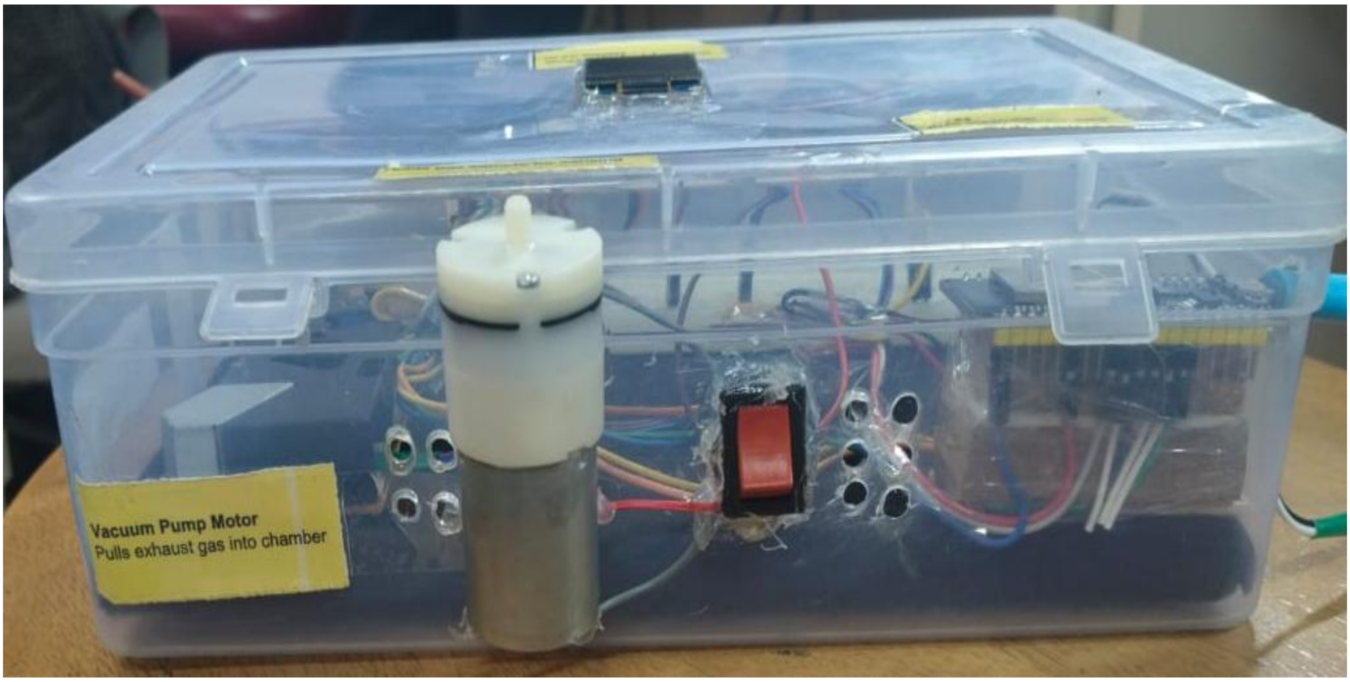

Figure 4.5.3 Vacuum Pump Assembly

Figure 4.5.4 Exhaust Gas Sampling Arrangement

4.6 OLED DISPLAY OUTPUT

The OLED display provides real-time local visualization of Carbon Monoxide (CO), gaseous pollutant concentration, particulate matter level, and system status for immediate monitoring.

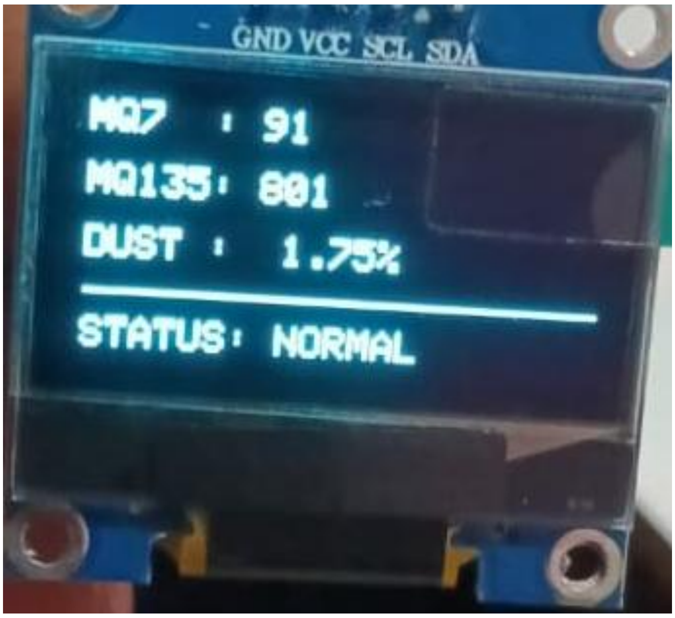

Figure 4.6.1.(a) OLED Display Output (With Sensors Identification)

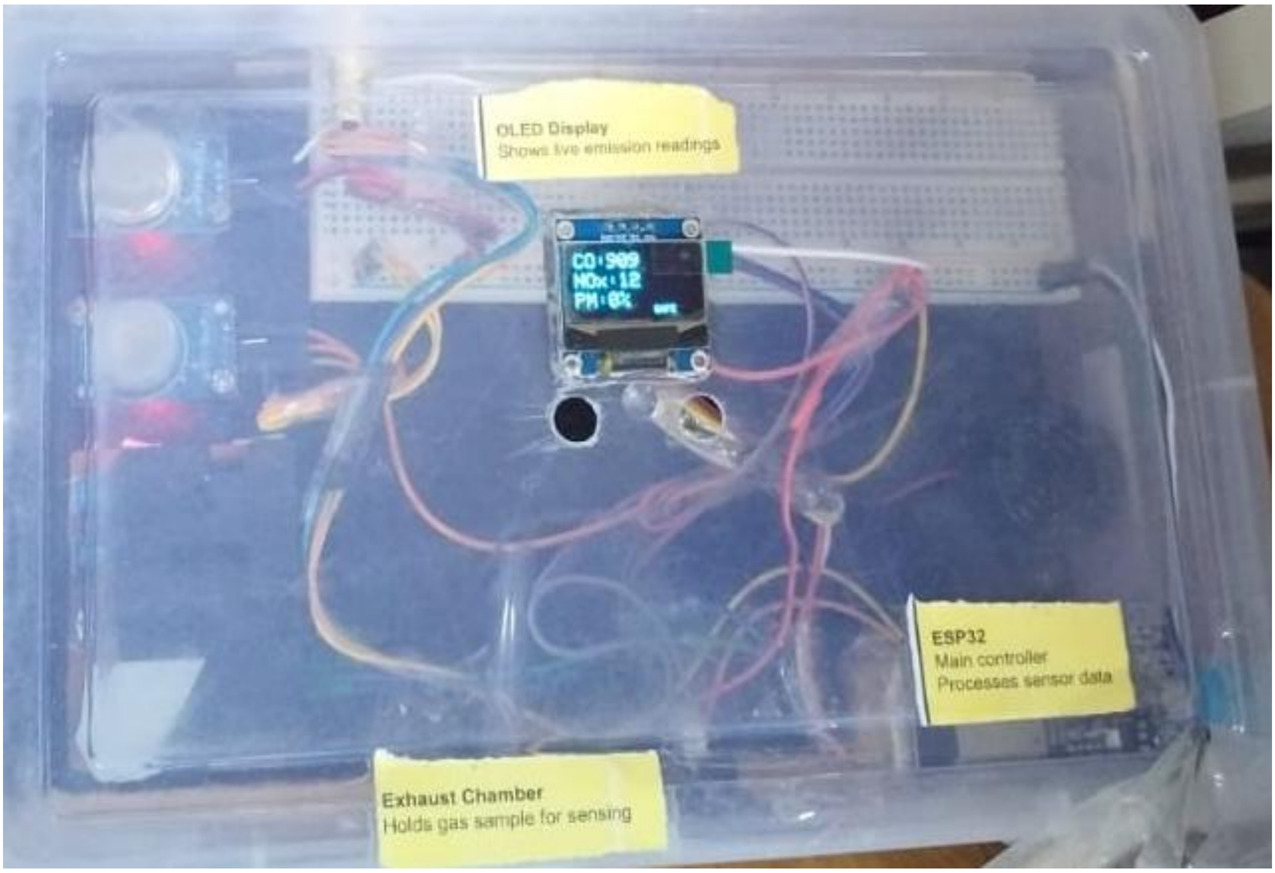

Figure 4.6.1.(b)OLED Display Output (With Gases Identification)

4.7 Cloud Monitoring Results

The ESP32 successfully transmitted sensor readings to the ThingSpeak cloud platform through Wi-Fi connectivity. The dashboard displayed graphical monitoring of CO, NOx, and particulate matter readings, enabling remote data monitoring and logging.

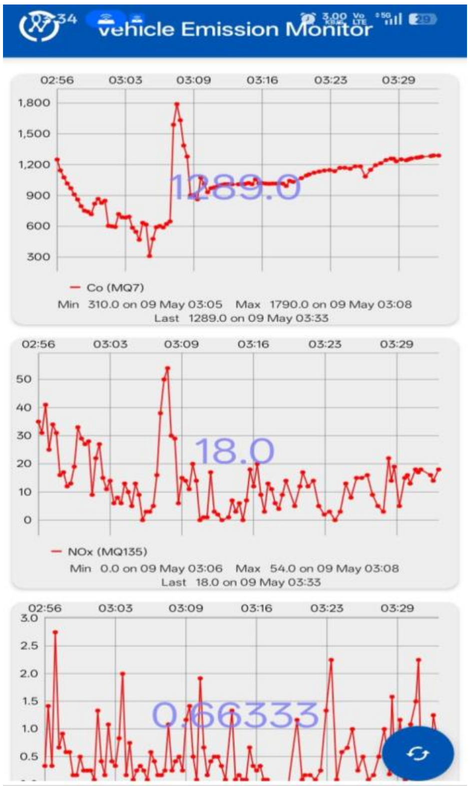

Figure 4.7.1.(A) ThingSpeak Monitoring Results

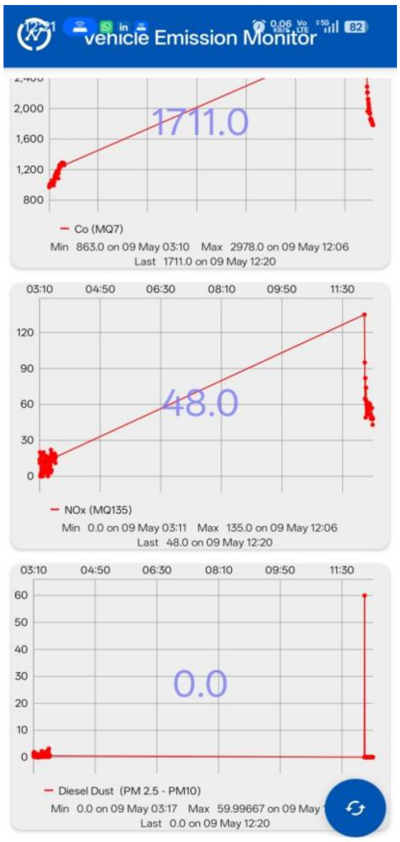

Figure 4.7.1.(B) Cloud Monitoring Output

4.8 EXPERIMENTAL RESULTS

The developed Live Emission Monitoring System prototype was experimentally tested under practical operating conditions to evaluate the performance of the integrated sensing and monitoring setup. Exhaust gas samples were drawn into the sensing chamber through the copper pipe inlet with assistance from the vacuum pump arrangement, ensuring continuous sampling of emitted gases.

During testing, the MQ-7 sensor successfully detected variations in Carbon Monoxide (CO) concentration depending on exhaust exposure conditions. The MQ-135 sensor responded to changes in harmful gaseous pollutant concentration, indicating variation in emission levels under different operating conditions. The DSM501B particulate matter sensor monitored dust and smoke concentration levels present in the sampled exhaust gas.

The OLED display provided real-time local visualization of all sensor readings, enabling immediate observation of emission levels. The warning indication mechanism also operated successfully, where predefined threshold conditions activated visual and audible alerts.

Simultaneously, the ESP32 microcontroller processed sensor data and transmitted readings to the ThingSpeak cloud platform through Wi-Fi connectivity. The graphical cloud monitoring results confirmed successful real-time remote monitoring and data logging capability.

The experimental observations confirmed successful integration and operation of sensing, processing, visualization, alert indication, and cloud communication modules within the developed prototype.

4.9 DISCUSSION

The experimental implementation of the proposed Live Emission Monitoring System demonstrated satisfactory performance for academic prototype and demonstration purposes. The integration of multiple sensors enabled simultaneous monitoring of Carbon Monoxide, gaseous pollutants, and particulate matter concentration within a single compact platform.

The use of ESP32 microcontroller provided efficient data acquisition, real-time processing, local display control, alert management, and cloud communication capability. The OLED display improved user accessibility by allowing direct observation of sensor outputs, while the ThingSpeak cloud platform enhanced monitoring by providing remote access to graphical emission data.

The vacuum pump assisted sampling mechanism improved exhaust gas flow toward the sensing chamber, resulting in better sensor exposure during testing. The warning alert mechanism further improved practical usability by providing immediate indication of unsafe emission conditions.

Although the developed system successfully demonstrates real-time monitoring functionality, sensor calibration accuracy can be further improved for industrial-grade precision. Environmental conditions such as temperature, humidity, and sensor aging may influence measurement consistency. However, as an academic low-cost prototype, the developed system effectively achieves its intended objective of demonstrating real-time vehicular emission monitoring using IoT technology.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 CONCLUSION

The proposed Live Emission Monitoring System for Petrol and Diesel Engines using IoT was successfully designed, developed, and experimentally implemented as a practical academic prototype for real-time vehicular emission monitoring. The primary objective of the project was to develop a compact, low-cost, and efficient system capable of continuously monitoring harmful vehicle exhaust emissions and providing both local and remote monitoring capability.

The developed system successfully integrated multiple sensing and embedded technologies into a single monitoring platform. MQ-7 sensor was used for detection of Carbon Monoxide (CO) concentration, MQ-135 sensor for monitoring gaseous pollutant variations, and DSM501B particulate matter sensor for dust and smoke concentration detection. The ESP32 microcontroller effectively handled sensor data acquisition, processing, threshold comparison, OLED display output, alert indication control, and wireless cloud communication.

The real-time OLED display provided immediate local monitoring of emission levels, while LED indicators and buzzer warning system enhanced practical usability by providing instant alert indication under moderate and high emission conditions. The integration of ThingSpeak cloud platform through Wi-Fi communication enabled continuous remote monitoring, graphical data visualization, and emission data logging.

The prototype was experimentally tested under practical exhaust gas sampling conditions using the developed copper pipe inlet and vacuum pump assisted gas sampling mechanism. Experimental observations confirmed successful detection of pollutant variation, reliable system response, and effective operation of the integrated sensing, display, alert, and cloud monitoring modules.

Although the developed system is intended as an academic prototype and not a certified industrial emission testing instrument, the project successfully demonstrates the practical implementation of IoT-based real-time vehicle emission monitoring using embedded sensing technology. The project effectively meets its intended objectives and provides a strong foundation for further development toward practical environmental monitoring applications

5.2 FUTURE SCOPE

The developed Live Emission Monitoring System provides a strong foundation for future improvements and practical enhancements. Although the present prototype successfully demonstrates real-time monitoring functionality, several advanced improvements can significantly enhance its accuracy, usability, and industrial applicability.

Future development can include integration of industrial-grade calibrated gas sensors for more accurate pollutant concentration measurement and compliance-level emission testing. Improved sensor calibration techniques can enhance consistency and measurement reliability under varying environmental conditions.

A dedicated mobile application can be developed for live remote monitoring, alert notifications, and historical emission trend analysis. Integration of GPS technology can further enable location-based vehicle emission tracking and fleet monitoring applications.

Advanced cloud analytics and database integration can support long-term emission data storage, predictive analysis, and maintenance recommendation systems. Artificial Intelligence and Machine Learning techniques may also be incorporated for emission behavior prediction and anomaly detection.

The system can be redesigned with a compact industrial enclosure, improved thermal protection, and robust sampling chamber for practical field deployment. Integration with government emission monitoring infrastructure or automated Pollution Under Control (PUC) systems can further enhance real-world applicability.

Thus, the proposed system offers significant future potential for development into a practical smart vehicular emission monitoring solution for environmental monitoring and pollution control applications.

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